

HW 8

1. Let $A \in \mathbb{R}^{n \times n}$ be SPD. By the Spectral Theorem, A is diagonalizable and has real eigenvalues. Let λ be an eigenvalue of A , such that $Av = \lambda v$ for $v \neq 0$. Using the definition of positive definite ($x^T A x > 0, x \neq 0$) and setting $x =$ eigenvector v ,

$$v^T A v > 0 \Rightarrow v^T (\lambda v) > 0 \Rightarrow \lambda (v^T v) = \lambda \|v\|^2 > 0$$

Since v is nonzero, then $\|v\|^2 > 0$, then dividing both sides gives

$$\lambda > 0$$

which completes the proof that the eigenvalues of A are strictly positive.

2. Let A, B be similar matrices. Let λ be any eigenvalue of B . Using the characteristic polynomial,

$$0 = \det(B - \lambda I) = \det(SAS^{-1} - \lambda I) = \det(SAS^{-1} - S(\lambda I)S^{-1}) = \det(S(A - \lambda I)S^{-1})$$

$$\hookrightarrow \lambda I = \lambda S S^{-1} = S(\lambda I)S^{-1}$$

$$= \det(S^{-1}) \det(A - \lambda I) \det(S) = \det(A - \lambda I) \cdot \frac{\det(S)}{\det(S)} = \det(A - \lambda I)$$

$\hookrightarrow \frac{1}{\det(S)}$

Conversely, using the same reasoning, if λ is any eigenvalue of A , then it is also an eigenvalue of B . Therefore, A and B share the same eigenvalues. Now, the eigenvectors can be characterized by

$$Bv = \lambda v \Rightarrow S^{-1}ASv = \lambda v \Rightarrow ASv = S\lambda v \Rightarrow ASv = \lambda Sv$$

This means that Sv is an eigenvector of A with the same eigenvalue λ while v is the corresponding eigenvector for matrix B .

$$3. \quad A = \begin{pmatrix} -1 & 2 & -3 \\ -3 & 4 & -3 \\ -1 & -2 & 1 \end{pmatrix}$$

$$\det(A - \lambda I) = 0 \Rightarrow \begin{vmatrix} -1-\lambda & 2 & -3 \\ -3 & 4-\lambda & -3 \\ -1 & -2 & 1-\lambda \end{vmatrix} = -(1+\lambda)((4-\lambda)(1-\lambda)-6) - 2(-3(1-\lambda)-3) - 3(6+(4-\lambda))$$

$$= -(1+\lambda)(\lambda^2-5\lambda-2) - 2(3\lambda-6) + 3\lambda-30$$

$$\lambda = 2 \quad (A - \lambda I)v = 0$$

$$\begin{pmatrix} -3 & 2 & -3 \\ -3 & 2 & -3 \\ -1 & -2 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0 \Rightarrow \begin{cases} -3x + 2y - 3z = 0 \\ -3x + 2y - 3z = 0 \\ -x - 2y - z = 0 \end{cases}$$

$$= -\lambda^3 + 4\lambda^2 + 4\lambda - 16 = 0$$

$$\Rightarrow \lambda^3 - 4\lambda^2 - 4\lambda + 16 = 0$$

$$\Rightarrow \begin{cases} -4x - 4z = 0 \\ 8y = 0 \end{cases} \Rightarrow \begin{cases} x = -z \\ y = 0 \end{cases} \quad v_1 = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}$$

$$\Rightarrow \lambda^2(\lambda-4) - 4(\lambda-4) = 0$$

$$\Rightarrow (\lambda^2-4)(\lambda-4) = 0$$

$$\begin{cases} -3x + 2y - 3z = 0 \\ 3x + 6y + 3z = 0 \end{cases}$$

$$\begin{cases} 3x + 6y + 3z = 0 \\ 3x + 6y - 3z = 0 \end{cases}$$

$$\Rightarrow (\lambda-2)(\lambda+2)(\lambda-4) = 0$$

$$\lambda = 2, -2, 4$$

$$\lambda = -2$$

$$\begin{pmatrix} 1 & 2 & -3 \\ -3 & 6 & -3 \\ -1 & -2 & 3 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0 \Rightarrow \begin{cases} x + 2y - 3z = 0 \\ -3x + 6y - 3z = 0 \\ -x - 2y + 3z = 0 \end{cases} \Rightarrow \begin{cases} 12y - 12z = 0 \\ -9x + 4y = 0 \end{cases} \Rightarrow \begin{cases} y = z \\ y = x \end{cases} \quad v_2 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

$$\lambda = 4$$

$$\begin{pmatrix} -5 & 2 & -3 \\ -3 & 0 & -3 \\ -1 & -2 & -3 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0 \Rightarrow \begin{cases} -5x + 2y - 3z = 0 \\ -3x - 3z = 0 \\ -x - 2y - 3z = 0 \end{cases} \Rightarrow \begin{cases} x = -z \\ 2x - 2y = 0 \end{cases} \Rightarrow \begin{cases} x = -z \\ x = y \end{cases} \quad v_3 = \begin{pmatrix} -1 \\ -1 \\ 1 \end{pmatrix}$$

Since eigenvectors corresponding to distinct eigenvalues are always linearly independent, then

A is diagonalizable.

$$A = VDV^{-1} = \begin{bmatrix} -1 & 1 & -1 \\ 0 & 1 & -1 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 2 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 4 \end{bmatrix} \begin{bmatrix} -1 & 1 & 0 \\ 1/2 & 0 & 1/2 \\ 1/2 & -1 & 1/2 \end{bmatrix}$$

$$V^{-1} = \frac{1}{\det(V)} \text{adj}(V)$$

$$\det(V) = -(1+1) - (-1) - (-1) = -2$$

$$\text{cof}(V) = \begin{pmatrix} 2 & -1 & -1 \\ -2 & 0 & 2 \\ 0 & -1 & -1 \end{pmatrix} \quad \text{adj}(V) = \begin{pmatrix} 2 & -2 & 0 \\ -1 & 0 & -1 \\ -1 & 2 & -1 \end{pmatrix}$$

4. spectral radius $\rho(A) = \max_{1 \leq i \leq n} |\lambda_i|$

a) verify that $\rho(A) \leq \|A\|$

Let λ be any eigenvalue of A with $\rho(A) = |\lambda|$ and let v be a corresponding eigenvector so $Av = \lambda v$. The matrix norm $\|\cdot\|$ induces

$$\frac{\|Av\|}{\|v\|} \leq \max_{v \neq 0} \frac{\|Av\|}{\|v\|} = \|A\| \Rightarrow \|Av\| \leq \|A\| \cdot \|v\| \Rightarrow \|\lambda v\| \leq \|A\| \cdot \|v\| \Rightarrow |\lambda| \cdot \|v\| \leq \|A\| \cdot \|v\|$$

$$\Rightarrow |\lambda| \leq \|A\| \Rightarrow \rho(A) \leq \|A\|$$

Since eigenvector $v \neq 0$, we can divide both sides by $\|v\|$ to get our final result.

b) Show that $\rho(A)$ is not a matrix norm by $\rho(A) = 0, A \neq 0$

$$A = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \quad \det(A - \lambda I) = \begin{vmatrix} -\lambda & 0 \\ 1 & -\lambda \end{vmatrix} = 0 \Rightarrow \lambda^2 = 0 \dots \lambda = 0 \text{ (mult. 2)}$$

$$\rho(A) = \max_{1 \leq i \leq 2} |\lambda_i| = \max(0, 0) = 0 \quad \therefore \text{violates } \|A\| = 0 \Leftrightarrow A = 0$$

By choosing $A = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$, we have chosen a nonzero matrix with $\rho(A) = 0$.

This makes $\rho(A)$ not a matrix norm.

5. Let $A \in \mathbb{R}^{n \times n}$ be invertible. Using SVD, show $\kappa_2(A^T A) = \kappa_2(A)^2$.

Let A have SVD: $A = U \Sigma V^T$,

$$A^T A = (U \Sigma V^T)^T U \Sigma V^T = V \Sigma^T \overset{I}{U^T U} \Sigma V^T = V \Sigma^T \Sigma V^T = V \Sigma^2 V^T = V \begin{bmatrix} \sigma_1^2 & & \\ & \sigma_2^2 & \\ & & \ddots \\ & & & \sigma_n^2 \end{bmatrix} V^T$$

Recall that any invertible matrix A with singular values $\sigma_1, \sigma_2, \dots, \sigma_n$ has $\kappa_2(A) = \frac{\sigma_1}{\sigma_n}$

Since $A^T A$ has singular values $\sigma_1^2, \sigma_2^2, \dots, \sigma_n^2$, $\kappa_2(A) = \frac{\sigma_1}{\sigma_n}$, so

$$\kappa_2(A^T A) = \frac{\sigma_1^2}{\sigma_n^2} = \left(\frac{\sigma_1}{\sigma_n} \right)^2 = \kappa_2(A)^2$$

6. a) Let $A \in \mathbb{R}^{n \times n}$ be diagonalizable w/ real eigenvals $\lambda_1, \dots, \lambda_n$ and real eigenvectors $v_1, \dots, v_n$.

Assume $|\lambda_1| \geq |\lambda_2| \geq \dots \geq |\lambda_n|$ and that $A \neq 0$, so at least $\lambda_1 \neq 0$.

Verify that $A^k x = \lambda_1^k \sum_{i=1}^n \alpha_i \left(\frac{\lambda_i}{\lambda_1}\right)^k v_i$

Given A is diagonalizable, it can be factored as $A = VDV^{-1}$, and because A

is diagonalizable, the eigenvectors form a basis for $\mathbb{R}^n$. For any $x = \alpha_1 v_1 + \dots + \alpha_n v_n \in \mathbb{R}^n$ where x is a linear combination of the eigenvectors $v_1, \dots, v_n$,

$$A^k x = (VDV^{-1})^k x = V D^k V^{-1} x = V D^k V^{-1} \sum_{i=1}^n \alpha_i v_i = V D^k \sum_{i=1}^n \alpha_i e_i = V \sum_{i=1}^n \alpha_i \lambda_i^k e_i$$

$$\hookrightarrow V e_i = v_i \dots e_i = V^{-1} v_i$$

$$\begin{bmatrix} v_1 & \dots & v_n \\ 1 & & 1 \end{bmatrix}^{-1} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$$

$$= \sum_{i=1}^n \alpha_i \lambda_i^k v_i = \lambda_1^k \sum_{i=1}^n \alpha_i \left(\frac{\lambda_i}{\lambda_1}\right)^k v_i$$

$\hookrightarrow$ factor out λ_1^k

b) describe $\lim_{k \rightarrow \infty} \frac{1}{\lambda_1^k} A^k x$ with $|\lambda_1| > |\lambda_2|$ and $\alpha_1 \neq 0$

$$\lim_{k \rightarrow \infty} \frac{1}{\lambda_1^k} A^k x = \lim_{k \rightarrow \infty} \frac{\lambda_1^k}{\lambda_1^k} \sum_{i=1}^n \alpha_i \left(\frac{\lambda_i}{\lambda_1}\right)^k v_i = \lim_{k \rightarrow \infty} \left[\alpha_1 \left(\frac{\lambda_1}{\lambda_1}\right)^k v_1 + \sum_{i=2}^n \alpha_i \left(\frac{\lambda_i}{\lambda_1}\right)^k v_i \right] = \boxed{\alpha_1 v_1}$$

$$\hookrightarrow \text{for } \lambda_i < \lambda_1 \Rightarrow \lim_{k \rightarrow \infty} \left(\frac{\lambda_i}{\lambda_1}\right)^k = 0$$

when $|\lambda_1| = |\lambda_2|$,

$$\lim_{k \rightarrow \infty} \frac{1}{\lambda_1^k} A^k x = \lim_{k \rightarrow \infty} \left[\alpha_1 \left(\frac{\lambda_1}{\lambda_1}\right)^k v_1 + \alpha_2 \left(\frac{\lambda_2}{\lambda_1}\right)^k v_2 + \sum_{i=3}^n \alpha_i \left(\frac{\lambda_i}{\lambda_1}\right)^k v_i \right] = \boxed{\alpha_1 v_1 + \sum_{\substack{i: |\lambda_i| = |\lambda_1| \\ 2 \leq i \leq n}} \alpha_i \left(\frac{\lambda_i}{\lambda_1}\right)^k v_i}$$

c) flop count for one iteration!

op flops

$A * v$ $2n^2 - n$

normalize v $2n$

Total: $\propto \boxed{2n^2 \text{ or } O(n^2)}$

d) Performance Analysis:

Random: converged in 37 iterations and returned the largest eigenvalue 4.

$x = (2 \ 1 \ 0)^T$: converged in 89 iterations, returned eigenvalue 4 (largest)

$x = (\sqrt{2} \ 0 \ -\sqrt{2})^T$: converged in 1 iteration, returned eigenvalue 2

it seems that a random vector remains in the "middle" in terms of performance and chooses the largest vector often.

e) To find the smallest eigenvalue λ_n using A^{-1} without forming A^{-1} explicitly, we can replace our matrix multiplication $A \cdot v$ with a function that solves the linear system $Au = v$. So, using `np.linalg.solve(A, v)`, the power method function on A^{-1} converges to the eigenvalue with the largest $|1/\lambda_i| = \min_{1 \leq i \leq n} |\lambda_i| = \lambda_n$.

```

import numpy as np

def power_method(A, x, tol=1e-10, max_iter=1000):
    v = x / np.linalg.norm(x)

    for i in range(max_iter):
        v_new = A @ v
        v_new = v_new / np.linalg.norm(v_new)

        # check for convergence
        if np.linalg.norm(v_new - v) < tol or np.linalg.norm(v_new + v) < tol:
            print(f"Converged in {i+1} iterations")
            break
        v = v_new
    else:
        print(f"Did not converge after {max_iter} iterations")

    u = A @ v_new
    i = np.argmax(np.abs(v_new))
    lam = u[i] / v_new[i]
    return v_new, lam

A = np.array([[-1, 2, -3],
               [-3, 4, -3],
               [-1, -2, 1]], dtype=float)

print("True eigenvalues:", np.sort(np.linalg.eigvals(A))[:-1])
print()

starting_vectors = {
    "Random": np.random.randn(3),
    "x=[2,1,0]": np.array([2., 1., 0.]),
    "x=[√2,0,-√2]": np.array([np.sqrt(2), 0., -np.sqrt(2)])
}

for name, x in starting_vectors.items():
    print(f"--- {name} ---")
    v, lam = power_method(A, x)
    print(f"Eigenvalue: {lam:.6f}, Eigenvector: {v}\n")

''' TERMINAL OUTPUT
True eigenvalues: [ 4.  2. -2.]

--- Random ---
Converged in 37 iterations
Eigenvalue: 4.000000, Eigenvector: [ 0.57735027  0.57735027 -0.57735027]

--- x=[2,1,0] ---
Converged in 89 iterations
Eigenvalue: 4.000000, Eigenvector: [-0.57735027 -0.57735027  0.57735027]

--- x=[√2,0,-√2] ---
Converged in 1 iterations
Eigenvalue: 2.000000, Eigenvector: [ 0.70710678  0.          -0.70710678]
'''

```